

COURSE OUTLINE



Basic Information:		
Course Code: CSE 311		
Course Title: Database Management System		
Credits: 3	Weekly: 3.00 hours	Total Week: 14

Prerequisite:

Course Code: CSE134 and CSE 221

Course Content (from syllabus):

Overview of database management systems; Relational Data Model; Database Design and Normalization; Data Flow Diagram, SQL Basics; Advanced SQL Queries; Database Implementation; Database Security; Data Warehousing and Data Mining; NoSQL Databases; Distributed Databases; Database Administration; Big Data and Cloud Databases; Emerging Trends in Databases

Course Description/Rationale:

DBMS refers to an introductory course for understanding the fundamental concepts, principles, and techniques of database management systems. DBMS aims at the design, implementation and maintenance of relational databases, as well as various database models, query languages, normalization techniques, and database administration.

Course Objective:

- Introduce students to core database concepts, including relational models, normalization, and basic SQL for designing and managing structured databases.
- Develop students' skills in implementing and optimizing relational databases using advanced SQL, indexing, and query optimization techniques.
- Prepare students to apply database security, distributed database concepts, and modern database tools to address real-world database management problems.

Course Outcomes (COs) with Mappings:

After completion of this course, students will be able to:

COs	CO Statements	POs	Learning Domains	Knowledge Profile	Complex Engineering Problem
CO1	Relate and apply fundamental database concepts including relational data models, normalization, and basic SQL to design and manage structured data systems.	PO1	C2, C3	K3 K4	EP1
CO2	Implement and optimize relational databases, incorporating advanced SQL queries, indexing techniques and query optimization strategies.	PO3	C3, C5, A2, A3, A4	K4 K5 K6	EP1, EP4
CO3	Apply and Analyze security techniques, distributed database architectures, and emerging tools and technologies in database management, using appropriate modern database tools while addressing real-world challenges.	PO5	C3, C4	K6	EP2

Bloom's Taxonomy	Knowledge Profile	CEP Attributes
Cognitive Domain	<i>K3: Engineering Fundamentals</i>	<i>EP1: Depth of Knowledge</i>
<i>C2: Understand</i>	<i>K4: Specialist Knowledge</i>	<i>EP2: Range of</i>
<i>C3: Apply</i>	<i>K5: Engineering Design</i>	<i>Conflicting</i>
<i>C4: Analyze</i>	<i>K6: Technology</i>	<i>Requirements</i>
<i>C5: Evaluate</i>	<i>K8: Emerging and Advanced Knowledge</i>	<i>EP4: Familiarity of issues</i>
Affective Domain		
A2: Responding		
A3: Valuing		
A4: Organizing		

Justification of CO-PO Mapping:

COs	POs	Justification of Mapping
CO1	PO1	CO1 focuses on fundamental database concepts, such as the relational data model, normalization, and SQL basics. This aligns with PO1, as it involves applying core engineering knowledge to solve database-related problems, ensuring a strong foundation for students in database management
CO2	PO3	CO2 involves designing, implementing, and optimizing relational databases using advanced SQL and indexing techniques. This matches PO3, which emphasizes the ability to design and develop engineering solutions to complex problems, ensuring students can create optimized, functional database systems.
CO3	PO5	CO3 covers security measures, distributed architectures, and emerging trends in database management. It aligns with PO5, as it focuses on using modern tools and technologies to address contemporary challenges in the field, ensuring students stay updated on the latest database advancements.

Justification of Knowledge Profile (KP) Mapping:

COs	Knowledge Profile (K)	Justification of Mapping
CO1	K3, K4	CO1 aligns with K3 as it builds foundational understanding of data representation, storage, and manipulation based on engineering fundamentals. It also maps to K4 as it involves applying specialized knowledge of relational modeling and database design principles essential to the discipline.
CO2	K4, K5, K6	CO2 maps to K4 for in-depth specialist knowledge in SQL optimization and database structures. It covers K5 through the design and development of optimized relational systems. K6 is addressed via practical application of indexing and performance tuning strategies.
CO3	K6, K8	CO3 maps to K6 through the application of modern database tools and technologies (e.g., NoSQL, Hadoop, cloud platforms) to solve practical problems, connecting theoretical concepts with real-world practice. It also addresses K8 by engaging students in the analysis of emerging trends, security practices, and distributed systems, encouraging research-oriented and lifelong learning in the field of database technologies.

Justification of Complex Engineering Problem (CEP) Attributes Mapping:

COs	CEP Attributes (EP)	Justification of Mapping
CO1	EP1	CO1 maps to EP1 because data originating from diverse domains may not fully satisfy ACID properties, resulting in depth of knowledge requires during normalization and database design.
CO2	EP1, EP4	CO2 addresses EP1 as solving complex problems necessitates crafting varied SQL queries to handle multiple technical challenges. It also relates to EP4, since students

COs	CEP Attributes (EP)	Justification of Mapping
		must tackle intricate issues such as atomicity and consistency while effectively communicating these technical complexities.
CO3	EP2	CO3 corresponds to EP2 because analyzing database security involves devising innovative solutions to rare and complex problems. Students explore conflicting requirements by studying the balance between security and performance during data retrieval processes.

Course Delivery Plan/Lesson Delivery Plan:

Weeks	Discussion Topic and Book Reference	Teaching Learning Activities (TLA)	COs	POs	Assessment Strategies for Topics
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Course outline discussion, Overview of database management systems, Evolution of databases	Lectures, basic SQL Practice	CO1	PO1	Midterm
	Lesson 2: Types of data in DBMS, Types of Language (DDL, DML, DCL, and TCL)				
Week-2 Lesson 3 & 4 [3 Hours]	Lesson 3: Relational model concepts Entity-Relationship (ER) modeling	Lectures, Worksheet, Group Practice in the classroom	CO1	PO1	Midterm
	Lesson 4: Relational model concepts Entity-Relationship (ER) modeling with Examples and exercises				
Week-3 Lesson 5 & 6 [3 Hours]	Lesson 5: Relational algebra basic and Concepts	Lectures, Practice test, case studies	CO1	PO1	Quiz-1, Midterm
	Lesson 6: Relational algebra’s Examples and exercise, Basic SQL syntax, Different types of Constraints.				
Week-4 Lesson 9 & 10 [3 Hours]	Lesson 7: Basic SQL syntax, Data definition language (DDL)	Lectures	CO1	PO1	Midterm
	Lesson 8: Data manipulation language (DML), Querying databases with SELECT				
Week-5 Lesson 11 & 12 [3 Hours]	Lesson 09: Basic of CRUD Operations with Examples and exercises	Advanced SQL Classwork, Optimization exercises, and Group presentations	CO2	PO3	Midterm
	Lesson 02: Basic of CRUD Operations with Examples and exercises				
Week-6 Lesson 13 & 14 [3 Hours]	Lesson 11: Subqueries, Set operations	Advanced SQL Classwork, Optimization exercises	CO2	PO3	Quiz-2, Midterm
	Lesson 12: Grouping and aggregation, Views and indexes				
Midterm Examination Week					
Week-7 Lesson 21 & 22 [3 Hours]	Lesson 13: Functional dependencies and normalization, Normal forms: 1NF, 2NF, 3NF	Lectures, SQL Practice	CO3	PO5	Final Term
	Lesson 14: Normal forms: 3NF, BCNF, Denormalization, Practical database design considerations				

Weeks	Discussion Topic and Book Reference	Teaching Learning Activities (TLA)	COs	POs	Assessment Strategies for Topics
Week-8 Lesson 15 & 16 [3 Hours]	Lesson 15: Join, Types of Joins, SQL on Join, Storage and file structures, Indexing techniques	Lecture, Classwork, and Group presentations	CO2	PO3	Quiz-3, Final Term
	Lesson 16: Query processing and optimization, Transaction processing				
Week-9 Lesson 17 & 18 [3 Hours]	Lesson 17: Security threats and vulnerabilities, Authentication and authorization	Lecture, research tasks, and discussions	CO3	PO5	Final Term
	Lesson 18: Encryption and access control Auditing and monitoring				
Week-10 Lesson 19 & 20 [3 Hours]	Lesson 19: Introduction to data warehousing, Data warehouse architecture	Lecture	CO1	PO1	Final Term
	Lesson 20: OLAP (Online Analytical Processing) and OLTP		CO3	PO5	
Week-11 Lesson 7 & 8 [3 Hours]	Lesson 21: Database administration tasks and responsibilities, Backup and recovery	Lectures, SQL Practice	CO1	PO1	Final Term
	Lesson 22: Performance tuning and optimization, Monitoring and troubleshooting				
Week-12 Lesson 23 & 24 [3 Hours]	Lesson 23: Introduction to distributed databases, Distributed database architecture	Lectures, Example SQL Exercise	CO1	PO1	Final Term
	Lesson 24: Replication and fragmentation, Distributed query processing and optimization		CO2	PO3	
Week-13 Lesson 25 & 25 [3 Hours]	Lesson 25: Overview of NoSQL databases, Types of NoSQL databases: document-oriented	SQL Classwork, Lectures	CO2	PO3	Final Term
	Lesson 26: Types of NoSQL databases: key-value, column-family, graph; Use cases and applications				
Week-14 Lesson 27 & 28 [3 Hours]	Lesson 27: Introduction to Cloud computing and databases Scalability and elasticity	Lecture	CO3	PO5	Final Term
	Lesson 28: Case studies and applications of Cloud Databases				
Final Examination Week					

PO Validation via Assessment:

Assessment Tools	Mapped COs	Mapped POs	Assessment/Tools
Midterm	CO1, CO2	PO1, PO3	Short Answers, SQL based queries (Basic and Subqueries), ERD diagram, Relational Algebra.
Final	CO1, CO2, CO3	P1, PO3, PO5	Normalization, Advanced SQL (Joins), Trigger, Transactions, Stored Procedure, Case Studies, Analytical Questions.

Assessment Pattern:

Assessment Tools	CO-wise Marks			Marks (Total=100)
	CO1	CO2	CO3	
Attendance	Not Applicable			07
Quiz	Not Applicable			15
Presentation	Not Applicable			08
Assignment	Not Applicable			05
Mid-Term	10	15		25
Final-Term	10	15	15	40
Total	25	30	10	100

CIE – Breakup (Theory) [60 marks]

Bloom's Criteria	Attendance	Class Test	Assignment	Presentation	Mid Exam
	07	15	05	08	25
Remember					
Understand		7			5
Apply		8			15
Analyze					5
Evaluate					
Create					

SEE – Semester End Examination [40 marks]

Bloom Criteria	Score for the Test
Remember	
Understand	
Apply	20
Analyze	10
Evaluate	10
Create	

Learning Materials: Textbook/Recommended Readings:

SL	Book Name	Author(s)	Publishers	Edition and year
1	Database System Concepts	Abraham Silberschatz, Henry F. Korth, S. Sudarshan	McGraw-Hill Education	7th Edition, 2019
2	Database Management Systems	Raghu Ramakrishnan, Johannes Gehrke	McGraw-Hill Education	4th Edition, 2018

Reference Books/Supplementary Readings:

SL	Book Name	Author(s)	Publishers	Edition and year
1	Database Design for Mere Mortals	Michael J. Hernandez	Addison-Wesley Professional	3rd Edition, 2013
2	SQL Queries for Mere Mortals	John L. Viescas, Michael J. Hernandez	Addison-Wesley Professional	3rd Edition, 2007
3	SQL Cookbook	Anthony Molinaro	O'Reilly Media	1st Edition, 2005

Prepared by Module Leader

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Signature